

```
1  #include<stdio.h>
2  #define MAX 3
3  int queue[MAX];
4  int front=-1, rear=-1;
5  void insert()
6  {
7      int value;
8      if(rear==MAX-1)
9      {
10         printf("Queue Overflow!\n");
11         return;
12     }
13     printf("Enter element to insert:");
14     scanf("%d", &value);
15     if(front== -1)
16         front=0;
17     rear++;
18     queue[rear]=value;
19     printf("%d inserted into the queue.\n", value);
20 }
21 void delete()
22 {
23     if(front== -1 || front > rear)
24     {
25         printf("Queue Underflow! (Empty Queue)\n");
26         return;
27     }
28     printf("%d deleted from the queue.\n", queue[front]);
29     front++;
30 }
31 void display()
32 {
33     if(front== -1)
34     {
35         printf("Queue is empty.\n");
36         return;
37     }
38     printf("Queue elements:");
39     for(int i=front; i<=rear; i++)
40     {
41         printf("%d ", queue[i]);
42     }
43     printf("\n");
44 }
45 int main()
46 {
47     int choice;
48     while(1)
49     {
50         printf("\n---Queue Menu---\n");
51         printf("1.Insert\n2.Delete\n3.Display\n4.Exit\n");
```

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```
14     scanf("%d",&value);
15     if(front==-1)
16         front=0;
17     rear++;
18     queue[rear]=value;
19     printf("%d inserted into the queue.\n",value);
20 }
21 void delete()
22 {
23     if(front==-1||front>rear)
24     {
25         printf("Queue Underflow! (Empty Queue)\n");
26         return;
27     }
28     printf("%d deleted from the queue.\n",queue[front]);
29     front++;
30 }
31 void display()
32 {
33     if(front==-1)
34     {
35         printf("Queue is empty.\n");
36         return;
37     }
38     printf("Queue elements:");
39     for(int i=front;i<=rear;i++)
40     {
41         printf("%d ",queue[i]);
42     }
43     printf("\n");
44 }
45 int main()
46 {
47     int choice;
48     while(1)
49     {
50         printf("\n---Queue Menu---\n");
51         printf("1.Insert\n2.Delete\n3.Display\n4.Exit\n");
52         printf("Enter your choice:");
53         scanf("%d",&choice);
54         switch(choice)
55         {
56             case 1:insert();break;
57             case 2:delete();break;
58             case 3:display();break;
59             case 4:return 0;
60             default:printf("Invalid choice!Try again.\n");
61         }
62     }
63 }
64 }
```

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"C:\Users\Admin\Desktop\18M24CS173\Linear queue.exe"

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:2
Queue Underflow!(Empty Queue)
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Queue is empty.
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:23
23 inserted into the queue.
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:45
45 inserted into the queue.
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:24
24 inserted into the queue.
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Queue Overflow!
```

```
---Queue Menu---
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Queue elements:23 45 24
```

```
---Queue Menu---
1.Insert
2.Delete
```

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"C:\Users\Admin\Desktop\18M24CS173\Linear queue.exe"

```
2.Delete
3.Display
4.Exit
Enter your choice:1
Enter element to insert:24
24 inserted into the queue.
```

---Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Queue Overflow!
```

---Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:3
Queue elements:23 45 24
```

---Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:2
23 deleted from the queue.
```

---Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:1
Queue Overflow!
```

---Queue Menu---

```
1.Insert
2.Delete
3.Display
4.Exit
Enter your choice:
```

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